

Unraveling Neutron Star Transients

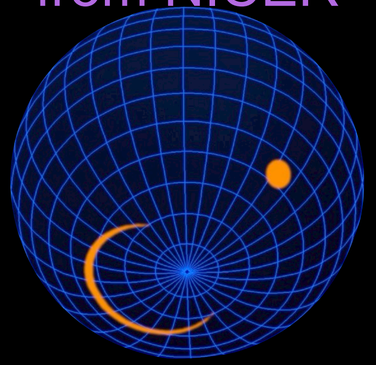
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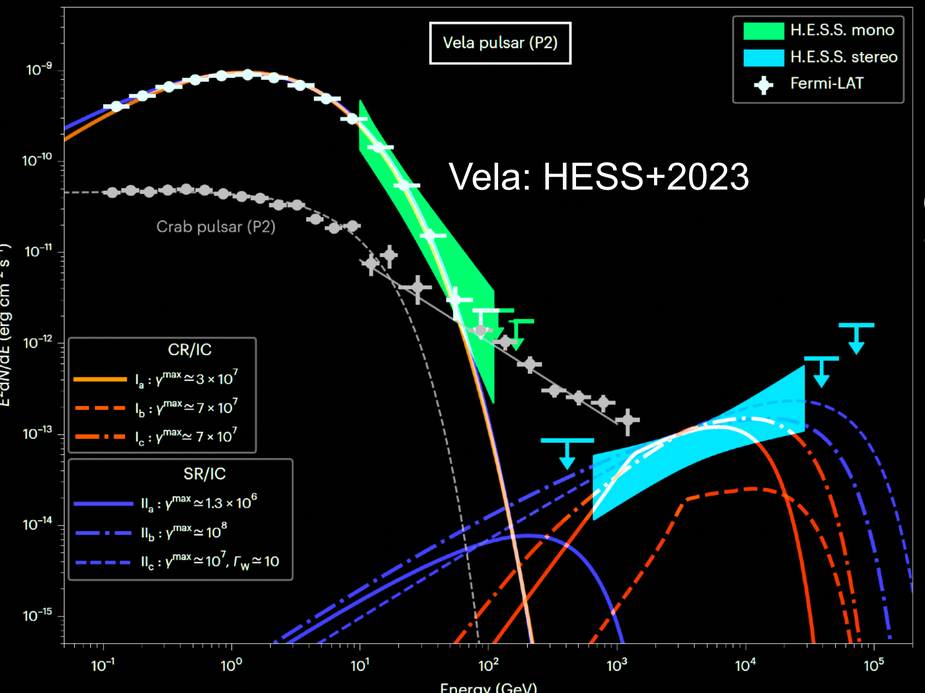
Neutron Stars: structure and emission

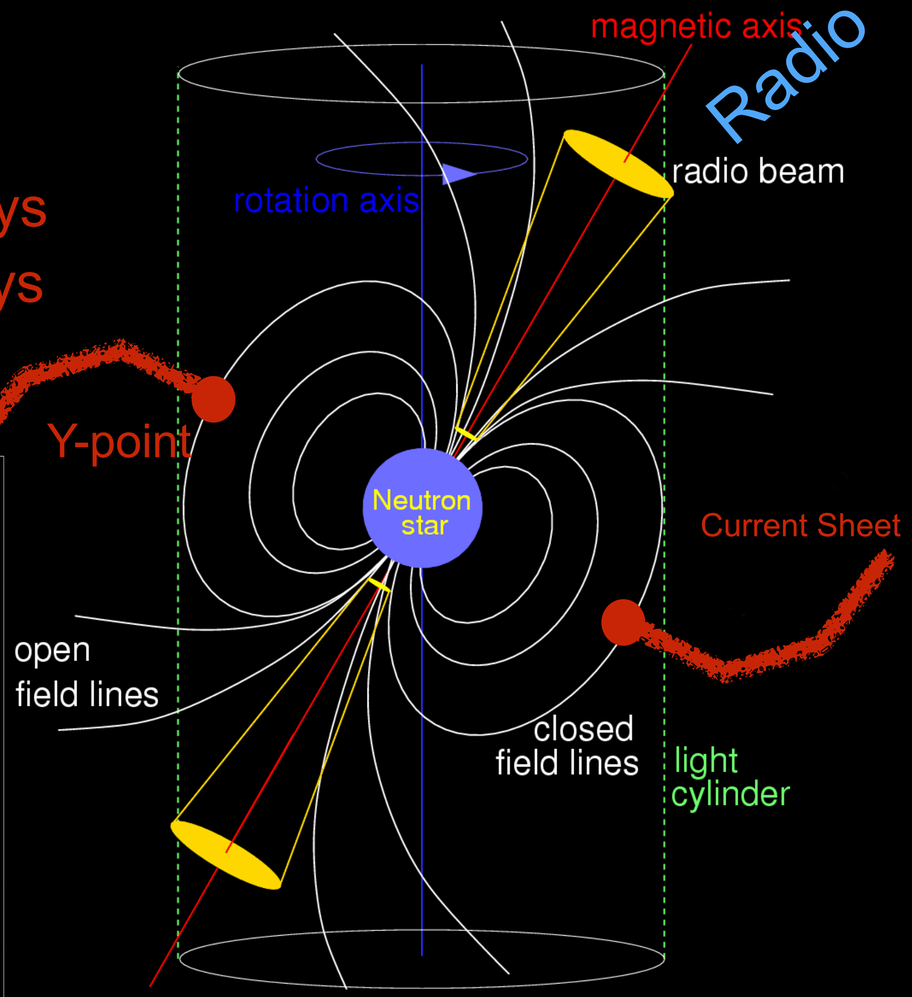
Thermal X-rays from NICER



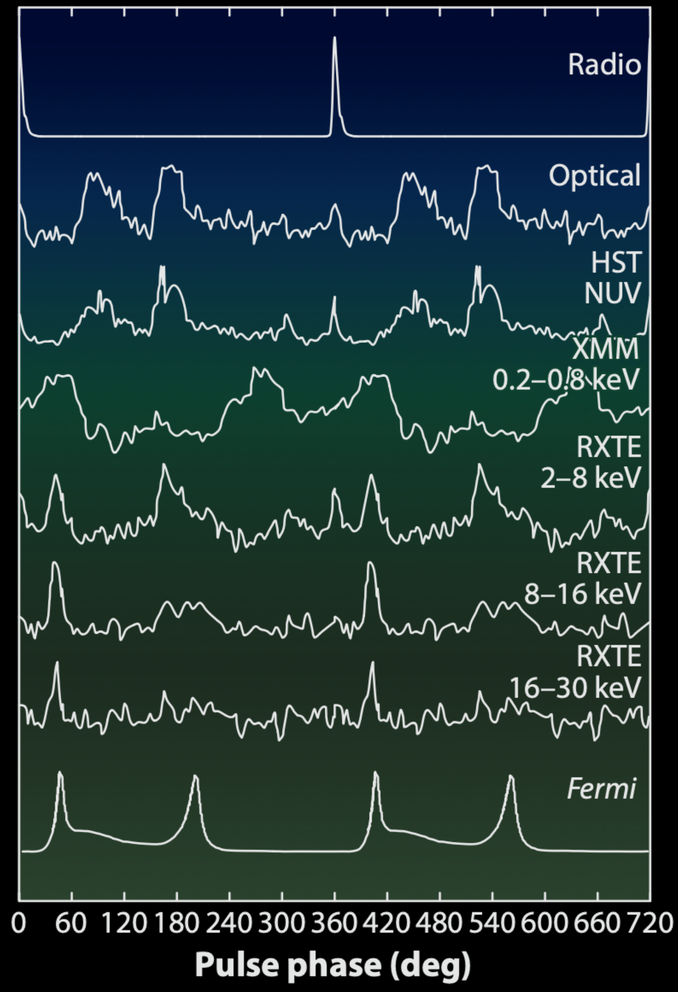
Optical, Xrays to TeV γ-rays

TeV emission from Pulsars





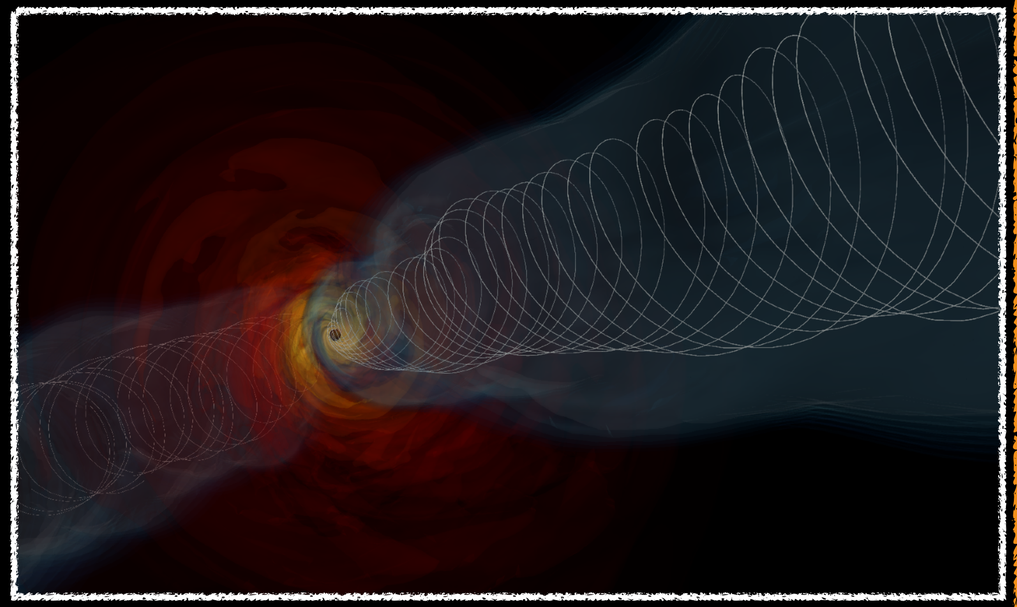
Vela pulsar*



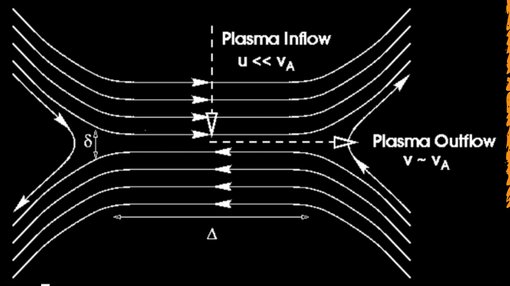
*Kuiper & Hermsen+15; Spolon+19, EPN, from Philippov & Kramer+22

Black Holes and Neutron Stars

Black Hole Accretion, Jets & Emission



Magnetic Reconnection



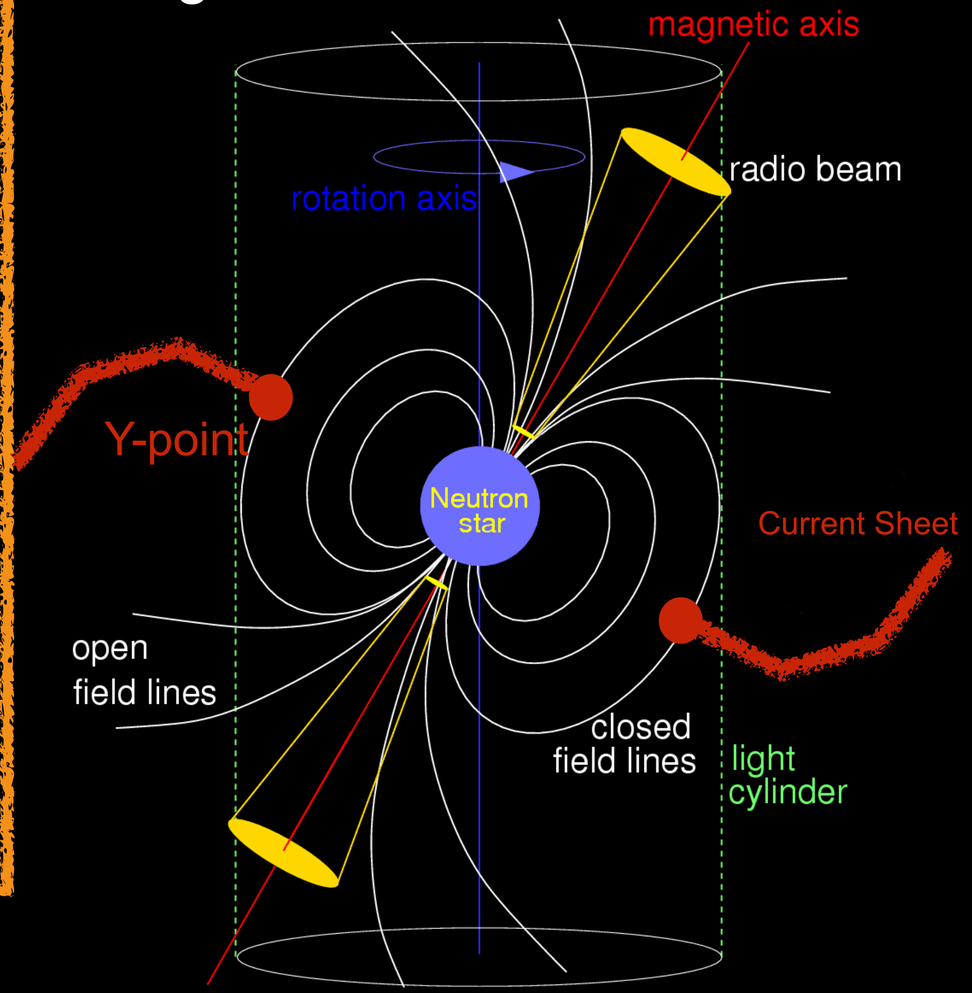
Commonalities

- (Non-)Spinning central object
- Current sheets
- Highly Magnetized regions, e.g., magnetospheres

$$\sigma \equiv \frac{B^2/4\pi}{\rho c^2} \gg 1$$

- Plasma interactions with dynamically strong B-fields

Isolated & Accreting NSs Magnetar Bursts & Pulsars



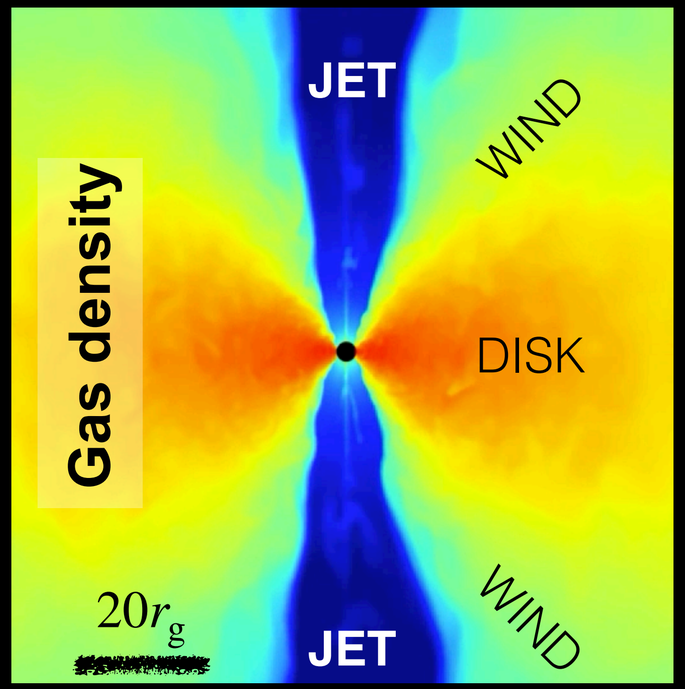
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Theory and Numerics

General Relativistic Magnetohydrodynamics (GRMHD)

- Ideal collisional fluid: global accretion dynamics
- Does not capture non-thermal/kinetic effects

H-AMR (Liska, KC+22)

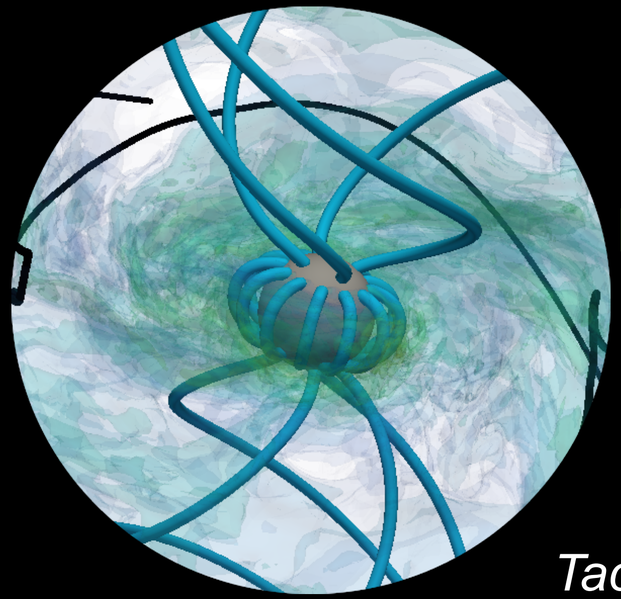


- GPU-accelerated
- Adaptive mesh refinement
- Local adaptive timestepping
- 2000^3 uniform grid accretion simulation on 6000 V100 GPUs *(Ripperda, Liska, KC+22)*

$$r_g \equiv \frac{GM}{c^2}$$

*Force-Free Electrodynamics (FFE)

- Highly magnetized plasma: jets, neutron stars
- Neglects plasma inertia and thermodynamics

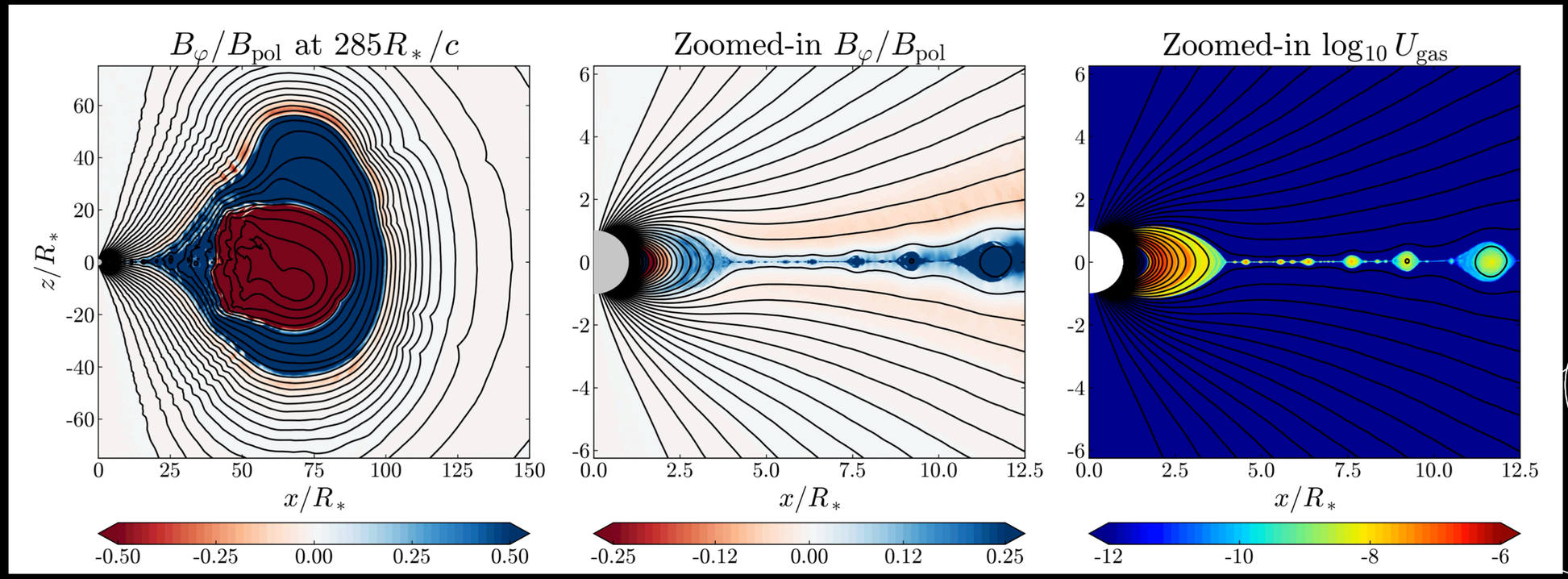


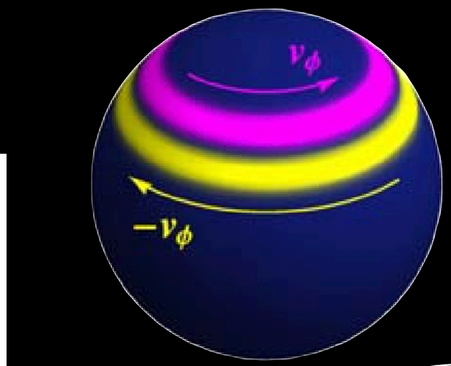
New Hybrid FFE-GRMHD with *H-AMR (KC+)*

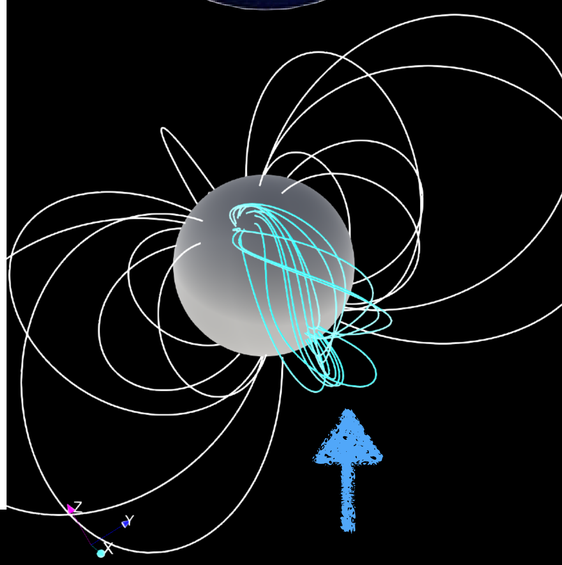
Tackling Neutron Star Magnetospheres and Accretion (3D pulsars on 3000 GPUs)

RETCO VI

Giant Flares from Magnetars







Twisted B-Field lines

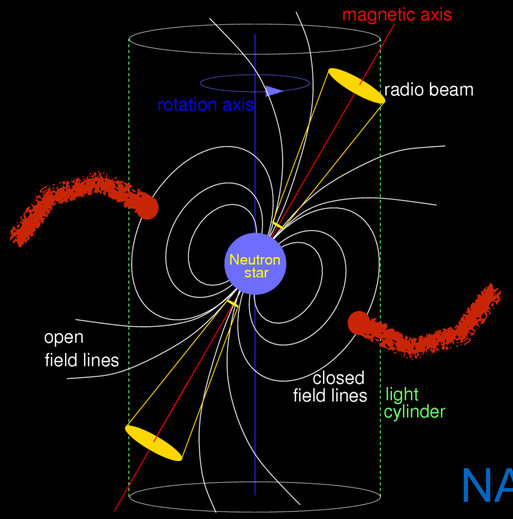
- Crustal motions (e.g., Magnetar quakes) power large eruptions:
  hot inner magnetosphere, bubble and a giant flare (~1-10% of E_{dip}~$10^{45\text{-}46}$ergs)

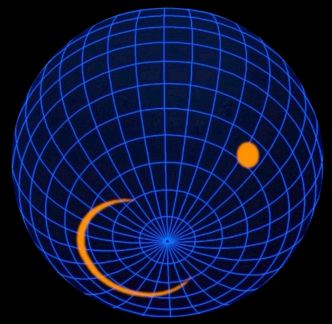
- Ejected bubble as a Fast Radio Burst (~10^{40}ergs) progenitor:
  via shocks (Beloborodov+23), interactions with wind/current sheet (Lyubarsky+20)

First MHD simulations of Magnetar Flares
KC, Philippov+
(in prep)
FFE: Mahlmann++, Yuan++

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Pulsar Magnetospheres

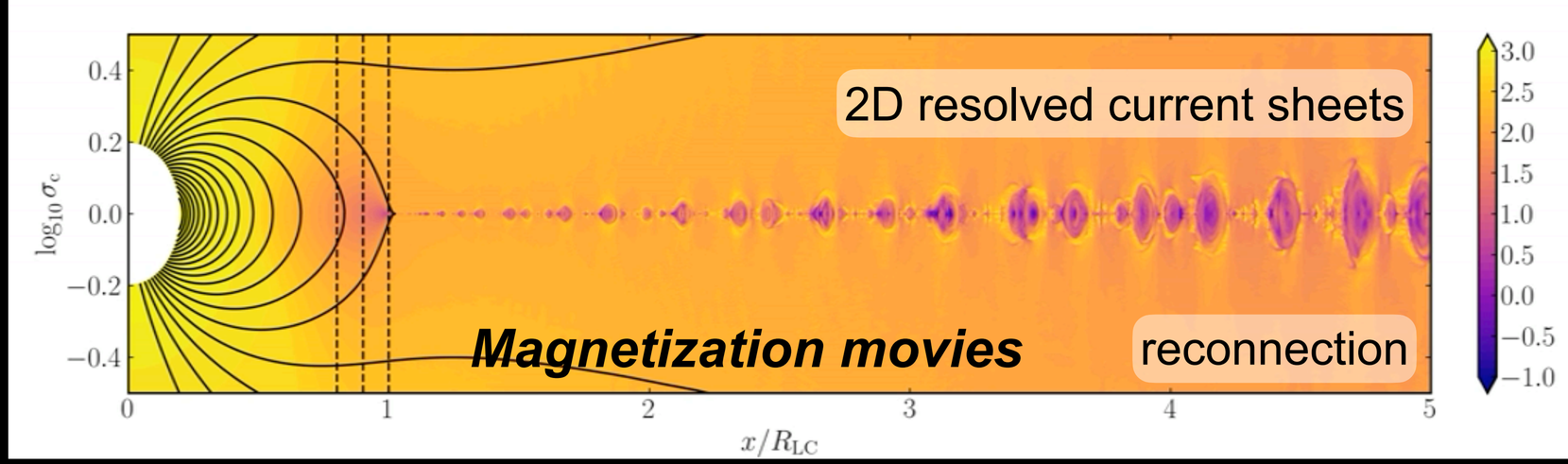




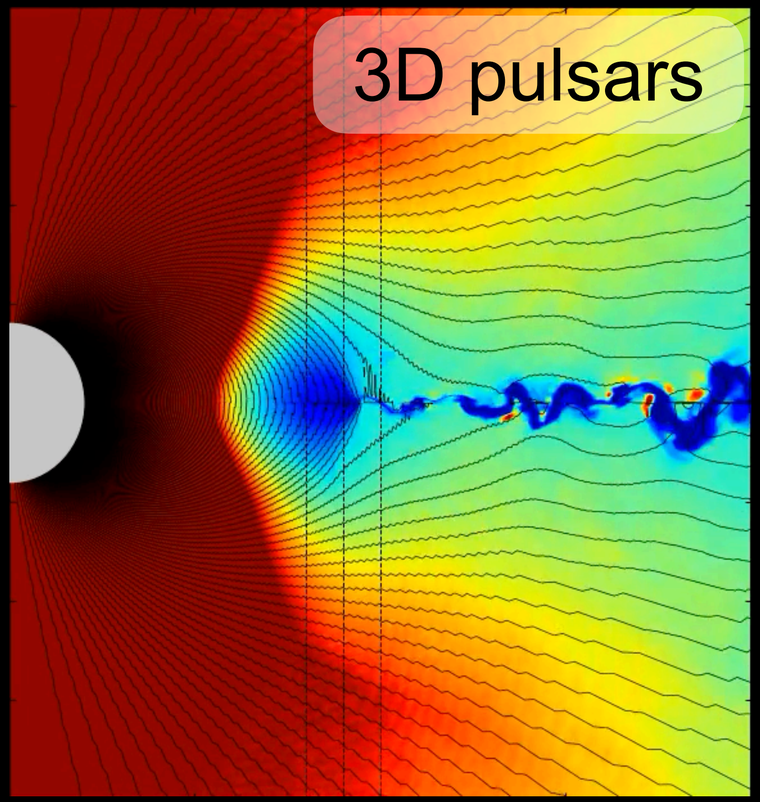
NASA NICER Pulsar surface map

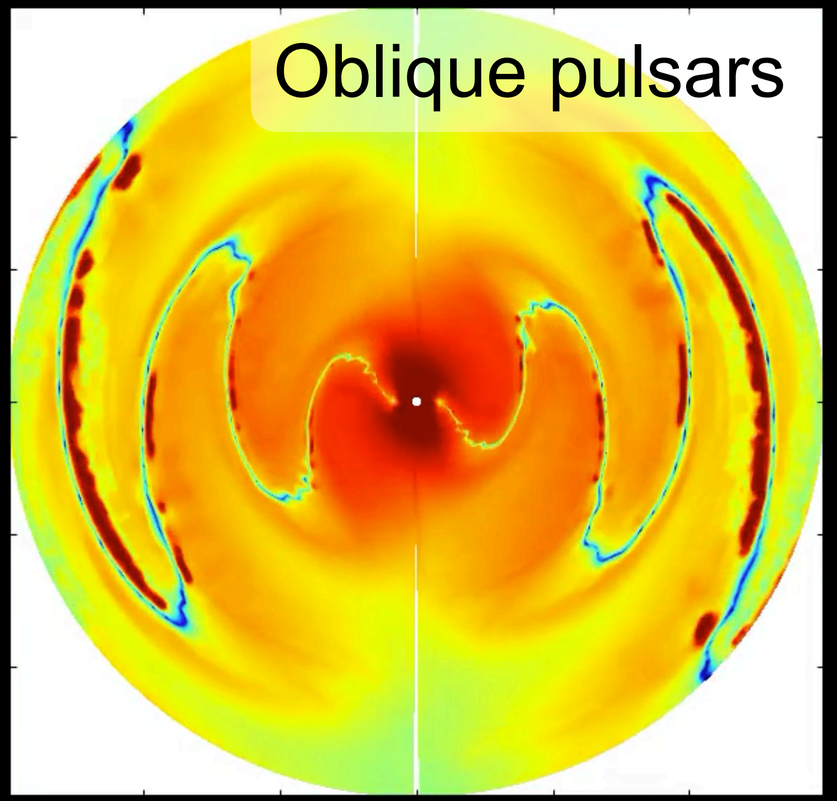
- current sheets, plasmoid, energetics, winds (striped)
- Tracking currents to the surface -> crustal heating
  (hotspot dynamics: Solanki, KC, Philippov+)

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First MHD sims of 3D Resolved Plasmoids in Pulsars
KC, Philippov+

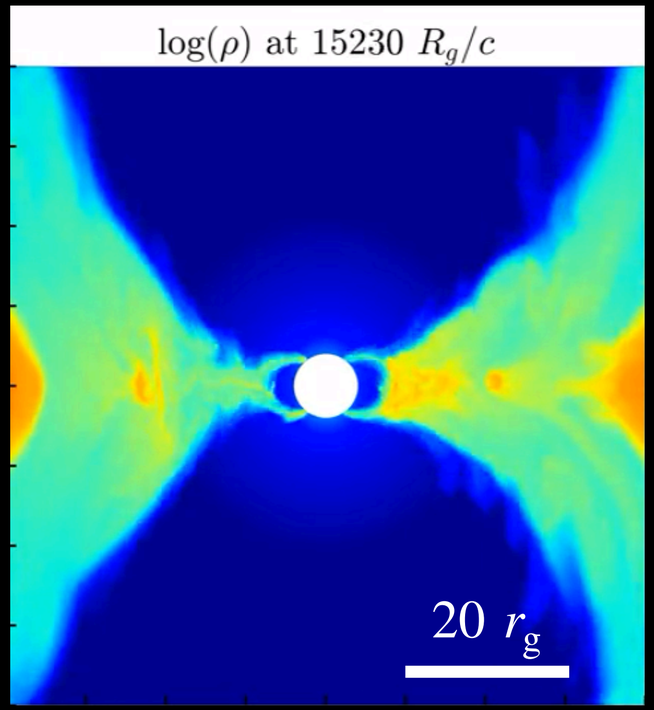


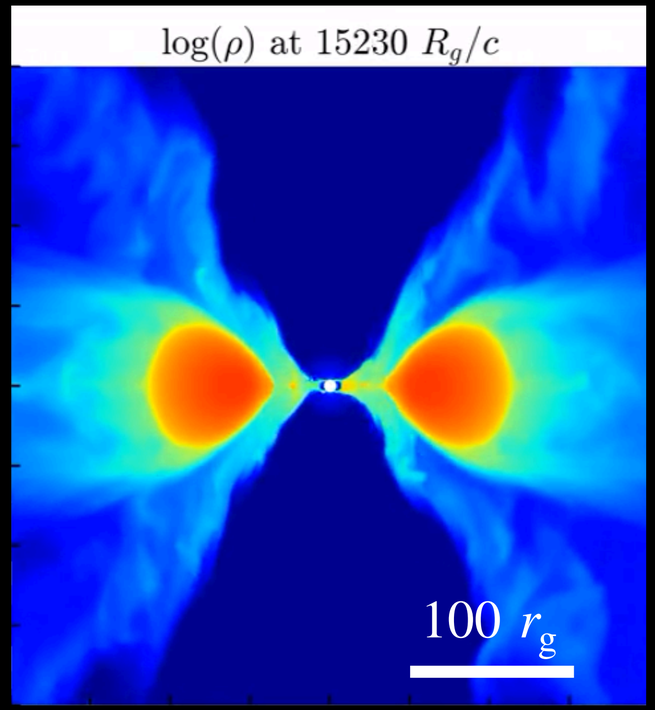


Accreting Pulsars

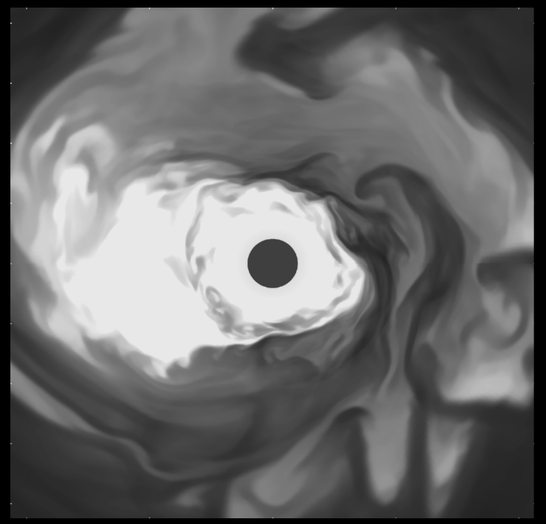
GRMHD simulations of accretion onto 3D pulsars

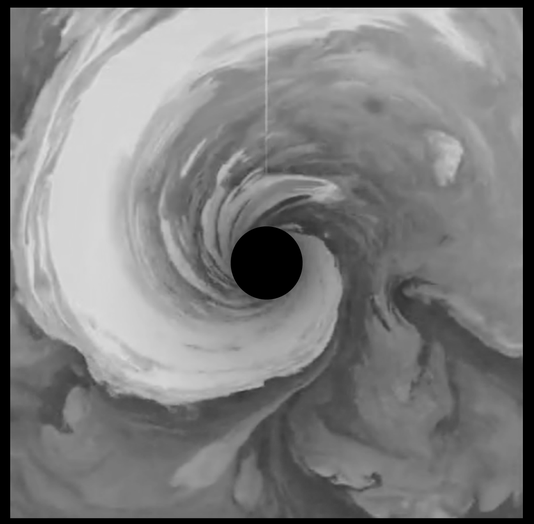
KC, Parfrey, Tchekhovskoy+ (in prep)





- Tracking accretion through columns
- Feedback on accretion from NS wind and pulsar current sheet





Magnetosphere-disk interactions in Pulsars (Left) and Black holes (Right)

Also see Parfrey & Tchekhovskoy 23

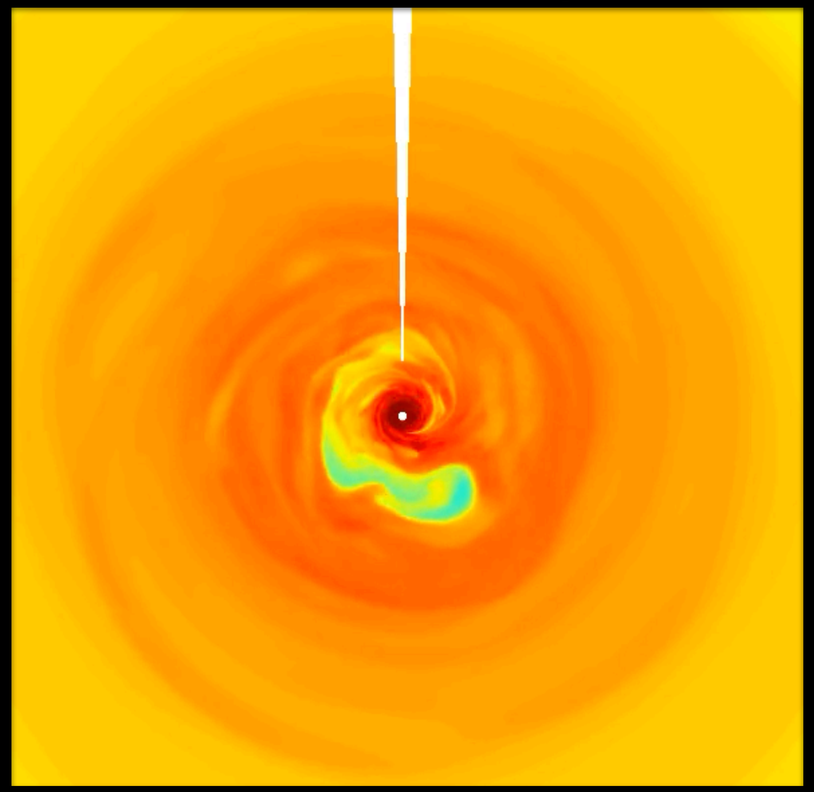
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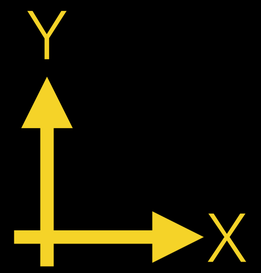
Accreting Pulsars: Magnetospheric feedback

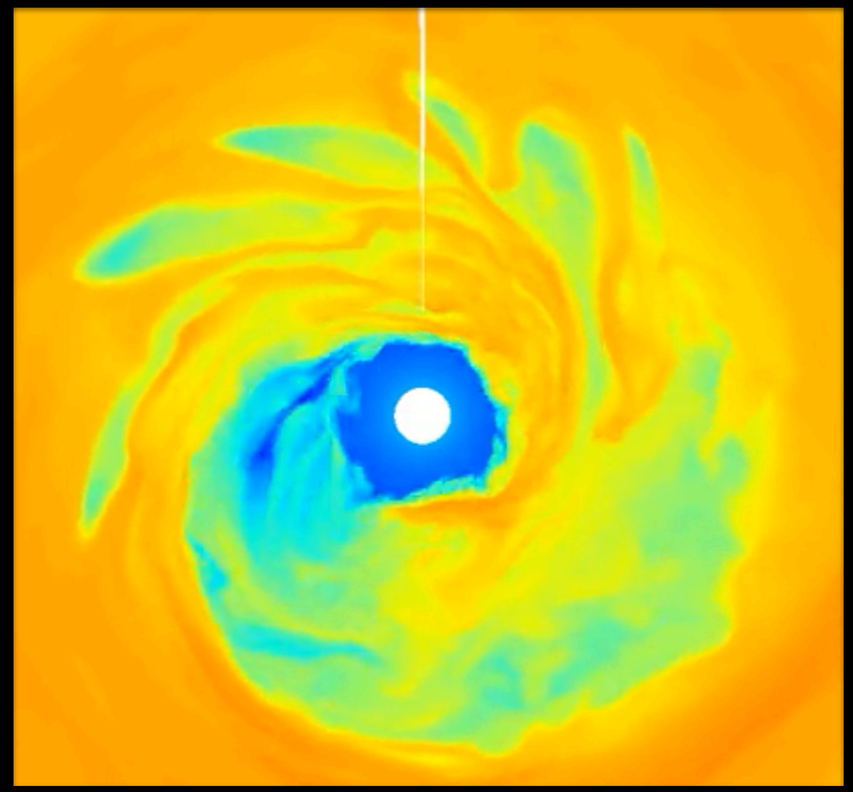
Black hole Eruptions

Neutron Star Arms

Equatorial cross-section







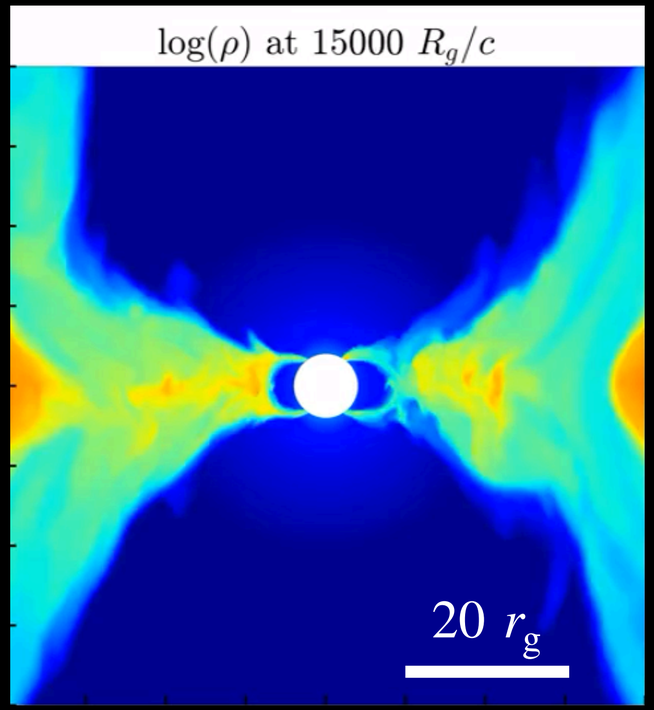
Bunched magnetic fields escape “randomly”

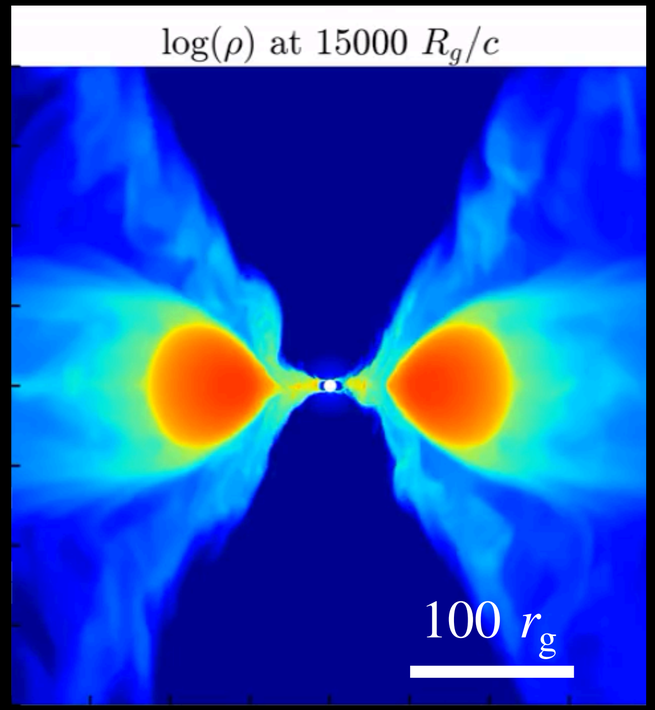
Part of the NS field punctures the disk and churns the gas at a rate slower than the NS rotation rate

Accreting Pulsars

GRMHD simulations of accretion onto 3D pulsars

KC, Parfrey, Tchekhovskoy+ (in prep)



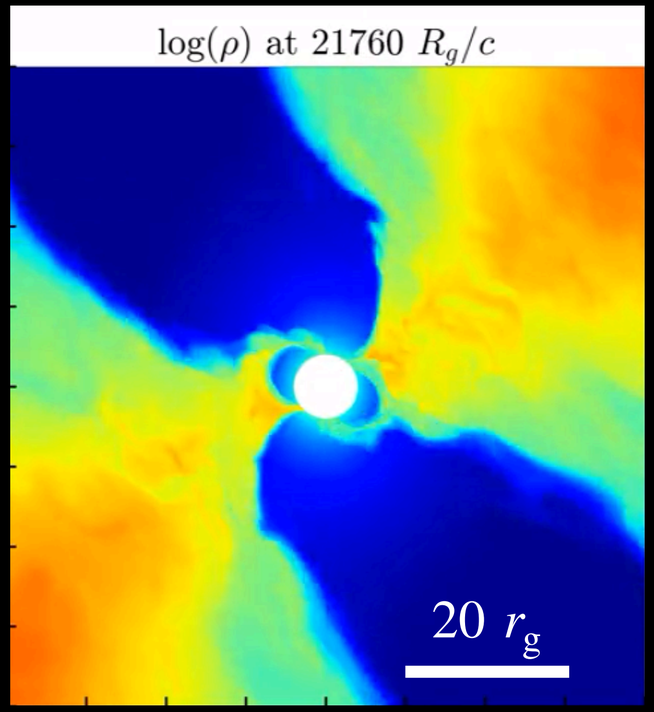


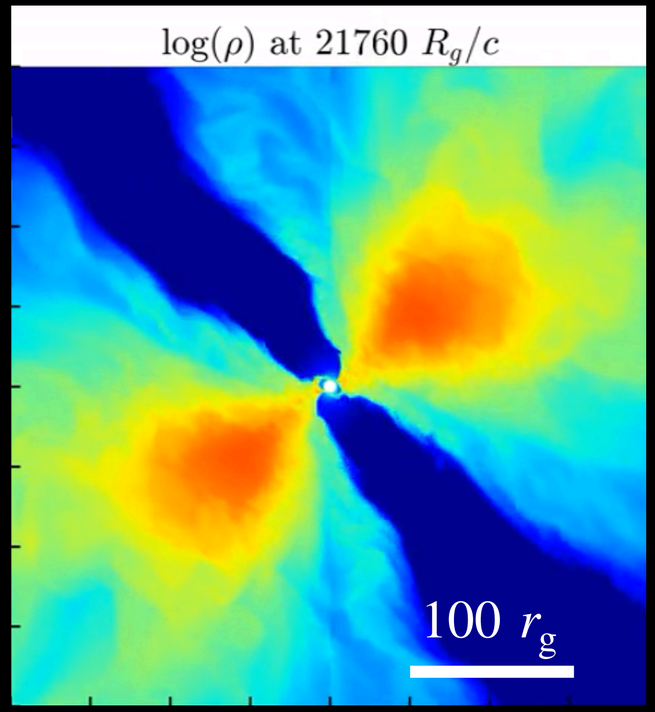
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Accreting Pulsars

GRMHD simulations of accretion onto 3D pulsars

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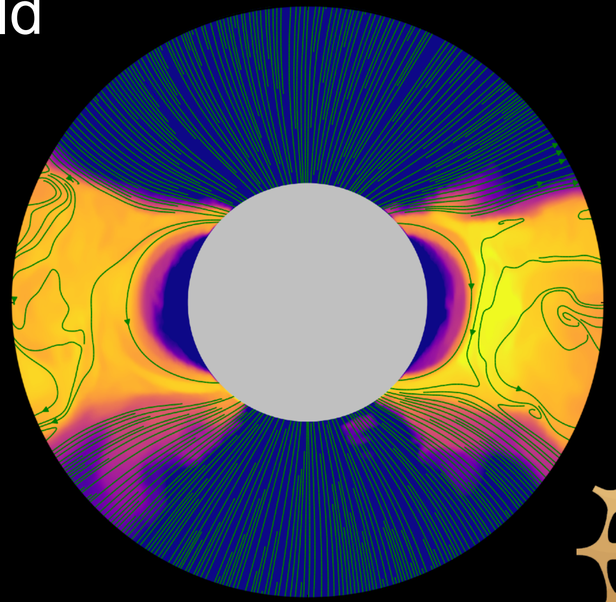
- Misaligned accretion of gas onto oblique pulsars
- Preliminary simulations indicate
  - Disk alignment due to strong NS field
  - Disk breaking can occur
  - Large variation in luminosities
  - Potential as a strong X-ray source

Summary: *New Frontiers in NS Magnetospheres*

- Isolated Pulsars show unstable current sheets and plasmoids that affects surface heating -> thermal X-ray signatures

- Strong magnetosphere-disk and wind-disk interactions in accreting pulsars hold interesting observational signatures and hold great potential as a source for QPOs

- Next steps involve self-consistent radiative physics for giant flares, current sheets and rapidly accreting neutron stars (ULXs)



ULX sources are thought to exhibit distinct signals from accretion columns

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